

WAIOTEKUMURAU (SANDS) FOREST MANAGEMENT PLAN (2024 – 2029)	
Index	1. Effective Date 2. Revision Date 3. Preamble and History 4. Location and Access 5. Topography & Climate 6. Geology and Soils 7. Previous Land Use 8. Vegetation 9. Management Objectives 10. Forest Resource 11. Forest Establishment 12. Future Operations 13. Forest Rotation 14. Forest Protection 15. Rare and Threatened Species 16. Environmental Impacts 17. Historic Sites and Wāhi Tapu 18. Rights of Other Parties 19. Recreation and Hunting 20. Special Issues 21. Legislation Key Work Programme (2024 – 2029) Appendix 1 – Forest Map Appendix 2 – Forestry Right Map
1. Effective Date	This management plan will run from 23/12/2024 until 22/12/2029
2. Next Revision Date	This plan will be updated 5 yearly with the next update scheduled for December 2029.
3. Preamble and History	This Forest Management Plan relates to a joint venture between Waimatenui Farms Limited (Grantor) and the Crown on the land and associated forests within the area shown in Appendix 1. The Forestry Right and Ancillary Deed were both executed on 23rd December 2019. Initially signed by the previous landowner Des Sands, the Forestry Right automatically transferred to Waimatenui Farms Limited when the sale and purchase of the underlying land was settled between the two parties (as per Ancillary Deed). Under the terms of the 35-year Forestry Right, the Crown meets all forestry costs, and pays an annual rental to the Grantor. There is no stumpage share agreement at harvest. Crown Forestry, a business unit within Te Uru Rākau, MPI, administers the Forestry Right on behalf of the Crown. Northland Forest Managers (1995) Limited has been engaged as the forest manager to oversee and manage day-to-day operations within the estate. This Management Plan should be read in conjunction with the Forestry Right.

4. Location and Access	The forest is located approximately 37 km south-west of Kaikohe, off Wai O Te Kumurau road and approximately 102 km from Northport at Marsden Point. There are two main 4WD metalled access tracks into the forest; one off Wai O Te Kumurau Rd and another along Waiowhata Rd, to the bottom fertiliser bin. They intersect and climb to the top fertiliser bin. In addition, there are several grass tracks throughout the block.																				
5. Topography & Climate	Topography is moderate to steep. Altitude varies from 100 to 530 metres above sea level. Climate records provide the following median annual statistics for this area; rainfall of 1500-1800 mm; temperatures of 13 degrees Celsius; 1850 hours of sunshine; and 20 days of soil moisture deficit per annum.																				
6. Geology & Soils	This forest grows on predominantly Waimatenui clay soils (YN, YNH). The erosion susceptibility classification of land in this forest under the National Environmental Standards for Commercial Forests (NES-CF) is approximately 40% green zone, 30% yellow zone, and 30% orange zone. Forestry operations are permitted activities in green and yellow zones, whilst some forestry operations in orange zone may require resource consent (such as earthworks).																				
7. Previous Land Use	The forest land was previously farmed with sheep and cattle.																				
8. Vegetation	Prior to planting, the land was predominantly grassland with some regenerating gorse and native scrub.																				
9. Management Objectives	The Forestry Right agreement outlines the objectives of the parties as follows: • to establish, manage, protect, store, carry away and sell the Trees; • use best endeavours to obtain a fully commercial return in respect of the objectives; • For the purpose of exercising such right and carrying out such duties, to have and enjoy such rights of access to and from the forest area, as are necessary to achieve the objectives. Under the Forestry Right agreement, the Grantee is deemed to have met its responsibilities if they act in compliance with this management plan. Crown Forestry reserves the right to revise this plan at its discretion, as provided for in the Forestry Right agreement.																				
10. Forest Resource	This <i>P.radiata</i> forest covers an area of approximately 497.4 hectares (ha) established in 2020 with a weighted average stocking of 986 stems per hectare (sph). In addition, there is 3.6ha planted outside legal boundary by Crown Forestry (Stand 1/04). Net stocked areas measured in January 2023 for each stand along with average planted stocking are listed in the table below: <table><tr><th>Cpt/Stand</th><th>YOE</th><th>Area (ha)</th><th>Average Planted Stocking (sph)</th></tr><tr><td>1/01</td><td>2020</td><td>108.7</td><td>936</td></tr><tr><td>1/02</td><td>2020</td><td>89.5</td><td>974</td></tr><tr><td>1/03</td><td>2020</td><td>299.2</td><td>1007</td></tr><tr><td>Total</td><td></td><td>497.4</td><td>986 (Wgt Avg)</td></tr></table> The above area statement is for Crown Forestry’s management purposes only. These areas should not be relied upon for any other purpose. Areas may change with remapping as trees and stocking gaps become visible to aerial imagery.	Cpt/Stand	YOE	Area (ha)	Average Planted Stocking (sph)	1/01	2020	108.7	936	1/02	2020	89.5	974	1/03	2020	299.2	1007	Total		497.4	986 (Wgt Avg)
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	There are two stands of older trees totalling approximately 7.5ha that are not part of the Forest Right, plus a number of younger stands planted by the Grantor on Road Reserve, which the Grantor is in the process of purchasing.																
11. Forest Establishment	Operations completed over the establishment phase of the forest are listed below: • Upgrade and maintenance of access tracks through the forest. • Erection of new forest signage. • Aerial desiccation of weeds and grass as land preparation • Ongoing pest control operations, primarily targeting goats. • 2020 - planting of 497.4 ha <i>P.radiata</i>, target stocking 1000 sph • 2021 - Blanking of areas with unacceptable mortality. • Aerial and spot releasing with herbicide. • Sail pruning of wind-affected areas. • Survival assessments following planting and blanking. • Foliage analysis and forest health surveys. • Erosion remediation (planting manuka on a slip face) • Aerial spraying to control Dothistroma.																
12. Future Operations	An unpruned (framing) regime was initially proposed for this forest. Crown Forestry have implemented a strategy to prune areas within some of the forests across our national estate. A range of factors have been considered, including but not limited to; the specific forest and region, tree growth rates and form, availability of labour and proximity to pruned log markets. This will provide benefits through; market flexibility and increased revenue at harvesting, and for the wider industry through extra work opportunities for silviculture contractors and increased potential pruned log supply for domestic sawmills. A review of the silvicultural strategy commenced in 2024, with pre-assessment data (PA) collected for 120 ha in stands 1/1 and 1/3. At the time of updating this management plan, the PA data was being collated and analysed by Northland Forest Managers. Initial results suggest that stand 1/3 is not suitable for pruning. The table below sets out the current tending scheduled over the next 5 years. The exact areas to be pruned, timing of operations and target pruned stocking will be determined from analysis of the pruning PA data. <table><tr><th>Year</th><th>Operation</th><th>Mean Top Height Range</th><th>Target Crop Sph</th></tr><tr><td>2025-2029</td><td>Prune (2 or 3 lifts)</td><td></td><td>TBC</td></tr><tr><td>2028-2029</td><td>Waste Thin (pruned stands)</td><td>12 – 14 m</td><td>As above</td></tr><tr><td>2026-2029</td><td>Waste Thin (unpruned stands)</td><td>12 – 14 m</td><td>500 (450 – 550)</td></tr></table> Crown Forestry reserves the right as forest owner to change regime.	Year	Operation	Mean Top Height Range	Target Crop Sph	2025-2029	Prune (2 or 3 lifts)		TBC	2028-2029	Waste Thin (pruned stands)	12 – 14 m	As above	2026-2029	Waste Thin (unpruned stands)	12 – 14 m	500 (450 – 550)
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13. Forest Rotation	Based on the regimes currently planned for this forest, a rotation length of between 25 and 30 years is expected; depending on tree growth and log markets. The size of this forest indicates clearfell harvesting will be spread over several years with road-line salvage operations likely to commence two years in advance of clearfell.																
14. Forest Protection	Pests The main pests present in the forest include goats, possums, and pigs. The scale of any pest control programme will be determined through regular inspections. Animal																

	damage is an ongoing risk in forests and control by shooting and trapping may continue to protect trees. The estate neighbours a large native reserve with a significant population of goats that will continue to encroach into the forest. External fences will be monitored to ensure breaches by livestock are limited. Biosecurity <i>Dothistroma</i> and Red Needle Cast diseases are becoming widespread in Northland, and close monitoring will identify whether treatment will be required. The Forest Biosecurity Surveillance scheme (FBS) established to replace the previous Forest Health Surveillance scheme reflects a new risk-based approach to surveillance. This focusses on new incursions rather than existing forest health pest and disease issues; and on protecting forest industry trade from biosecurity threats. This is partly funded by the Forest Growers Levy. Forest Owners Association and the Farm Forestry Association have jointly engaged SPS Biosecurity (SPS) as the provider of the surveillance programme. Frequency and intensity of health surveys will vary from forest to forest dependant on the risk profile of the estate. Forest owners may directly negotiate with SPS for the provision of a health survey over and above the sites surveyed under the FBS programme. Any detection of unexplained forest health issues should be reported as soon as possible to Crown Forestry and SPS (http://www.spsbiota.co.nz). Weeds Common weeds present within the forest include gorse, kikuyu grass and various pasture weeds. The Northland Regional Pest Management Plan (2017-2027) will be used as a guide to ensure weed control requirements are met. Fire Crown Forestry and Northland Forest Managers (1995) Ltd will engage with Fire and Emergency NZ (FENZ), with operational procedures documented within the Regional Fire Response Plan. Maps of the forest identifying location, access points and tracks, water points, and topography, along with contact details for landowners and the forest manager have been provided to FENZ. In the event of a fire, a '111' call will activate a response from FENZ. Forest Security Security requirements will be monitored and reviewed periodically. The requirement for forest security will be assessed based on the impact to both the tree crop; and health and safety of our workers/visitors. Signage will be considered and erected if appropriate.
15. Rare and Threatened Species	If any rare, threatened, or endangered species are identified within the estate, management plans will be prepared to ensure active management takes place. Industry protocols (http://rarespecies.nzfoa.org.nz/) will be followed to limit the impact of forest operations on the habitats of these species.
16. Environmental Impacts	Forest operations will adhere to relevant Northland Regional Council and Far North District Council rules and plans, the NZ Forest Accord, and the National Environment Standards for Commercial Forestry. Spot spray releasing will be preferred over aerial releasing on erosion prone or sensitive sites, when target weed species, topography, and labour availability permit. <u>Key resources:</u> NZFOA Environmental Code of Practice: https://www.nzfoa.org.nz/resources/file-libraries-resources/codes-of-practice/44-environmental-code-of-practice/file

	NZFOA Forest Practice Guides: https://docs.nzfoa.org.nz/forest-practice-guides/ National Environmental Standards for Commercial Forestry (NES-CF): National environmental standards for commercial forestry Ministry for the Environment Comply with the “Biosecurity (National PA Pest Management Plan) Order 2022” as regards to Kauri dieback: Biosecurity (National PA Pest Management Plan) Order 2022 (SL 2022/208) – New Zealand Legislation
17. Historic Sites and Wāhi Tapu	No historic or wāhi tapu sites have been identified to date in the forest area. Protocols are in place in the event of any accidental discovery. A desktop exercise was undertaken by Northern Archaeological Research Ltd, in June 2023. Their findings were that: • No archaeological sites have been recorded in the forest area. • However, the recorded archaeological landscape in the vicinity of the forest, suggests that there is a likelihood for a small number of unrecorded archaeological sites to be located within the Waiotekumurau block, likely above the river flats and possibly on high points. It was recommended that select areas of the Waiotekumurau Forest Block be surveyed for archaeological sites. It was also noted that NRC LIDAR information could be reviewed prior to any survey to aid in selecting areas for the potential for archaeological or historic remains.
18. Rights of Other Parties	Other Contractors & Visitors There will be other contractors and visitors entering the estate for various purposes. Forestry contractors will fall under the management of the forest manager and their health & safety system. Other contractors and visitors will be directly managed by the Grantor but only after obtaining prior approval from, and meeting all the requirements of, the forest manager.
19. Recreation and Hunting	There is currently no access for recreation and hunting within the forest area.
20. Special Issues	Grazing No grazing is currently allowed within the Forestry Right area. The Grantor is required to maintain the external fences so that livestock do not encroach the planted areas. Potential subdivision In an event the underlying land is being subdivided, Crown Forestry’s access to the forest needs to be protected to allow general forest management and future harvest operations to continue unimpeded. Crown Forestry should be consulted prior to any subdivision, so any potential adverse effects can be discussed and mitigated. A non-objection covenant will also need to be to be signed off by any new owner of the subdivided land. Landowner’s own planting (outside of Forestry Right Area) The Grantor has recently planted areas outside of the current agreed Forestry Right Area (i.e. powerline corridor and paper road that is in the process of being purchased). These areas are not under Crown Forestry management and are solely the responsibility of the Grantor.

	Grantor Participation in the ETS The Grantor may apply to register as a participant in the NZ ETS in relation to their interest in the land. If the Grantor does decide to apply to enter the NZ ETS they will bear any and all of the risks related to this and are solely responsible for any obligations under the Climate Change Response Act (2002) (“the CCRA”). Crown Forestry manages the forest to achieve the best commercial return from the sale of forest produce and reserves the right to adopt or vary its management in response to market conditions and forest growth. Management of a forest to maximise forest produce return may not necessarily align with optimum management for carbon revenue.
22. Legislation	1. Health & Safety at Work Act 2015 The Health & Safety at Work Act (HSWA) 2015 is New Zealand’s workplace health and safety law. Landowners, forestry right holders and forest management contractors and subcontractors are all classified as Persons Conducting a Business or Undertaking (PCBU) under the HSWA. All PCBU’s have a duty of care to people’s health and safety at work and are required to consult, co-operate, and coordinate to undertake all reasonably practicable steps to ensure a safe workplace is achieved. Those in control of workplaces are required to assign health and safety responsibilities as appropriate and ensure accountability. Training must be provided at all levels for people to effectively meet these responsibilities. These provisions are normally set out in the form of a company or contractor Health & Safety Policy and monitored through Health & Safety systems. Crown Forestry as Forest owners insist on strict compliance with company Policies, Procedures and Rules. All contractors must provide their own Health & Safety Policy and demonstrate consistent performance of their responsibilities, in line with Crown Forestry’s requirements. Crown Forestry, Northland Forest Managers Ltd, and Waimatenui Farms Ltd all place significant importance on health and safety and strongly support measures to ensure the safety and wellbeing of workers, contractors, and visitors within forest estates. They will work together to comply with the HSWA and identify and promote initiatives to improve the health and safety and wellbeing of people working in or visiting the forest under their respective purview. https://worksafe.govt.nz/managing-health-and-safety/getting-started/health-and-safety-at-work-quick-reference-guide/#lf-doc-43983 2. Resource Management Act 1991 The Resource Management Act 1991 promotes sustainable management of natural and physical resources. It is the principal statute for the management of land, water, soil, and other resources. The provisions of the Act are administered by Councils through District and Regional Plans which set out the policies and rules in the area. Where central Government wishes to provide local Councils with direction on environmental issues it can issue a National Environmental Standard. https://www.mfe.govt.nz/rma 3. National Environmental Standards for Commercial Forestry (NES-CF) The National Environmental Standards for Plantation Forestry (NES-PF) regulations, administered under the RMA, were released in 2018. They provide nationally consistent regulations to manage the environmental effects of plantation forestry. A 12-month review took place in 2020 and following further consultation and recommended amendments made following the Ministerial Inquiry into Land Use (MILU), the NES-CF (Commercial Forestry) became effective from 3rd November 2023. These new standards will apply to both plantation forests and continuous-cover forests (carbon forests) that are deliberately established for commercial purposes.

These standards have introduced new requirements for forestry, particularly for replanting, afforestation, and harvesting operations. They will also:

- enable councils to consider more factors when making rules about forestry in their plans, including its location.
- require carbon foresters to plan out how they will meet environmental requirements for different forestry activities on their sites.
- state clear rules for carbon forest harvest should this be undertaken.
- introduce a range of operational changes including a new permitted activity standard for managing forestry slash at harvest and new requirements around management of wilding trees.

The NES-CF is based on good forestry practice and covers eight core forestry activities:

- Afforestation
- Pruning and thinning to waste
- Earthworks
- River crossings
- Forestry quarrying
- Harvesting
- Mechanical land preparation
- Replanting.

Forest operations are deemed permitted or restricted depending on the erosion susceptibility classification of the forest land. This is colour coded under four zones as described below:

- Green zone: low erosion risk
- Yellow zone: medium erosion risk
- Orange zone: high erosion risk; and
- Red zone: very high erosion risk

Other science-based tools used under the NES-CF regulations to assist councils and foresters plan and manage forestry operations are:

- Fish Spawning indicator
- Wilding Tree Risk Indicator

<https://www.mpi.govt.nz/growing-and-harvesting/forestry/national-environmental-standards-for-plantation-forestry/>

4. Other Relevant Legislation

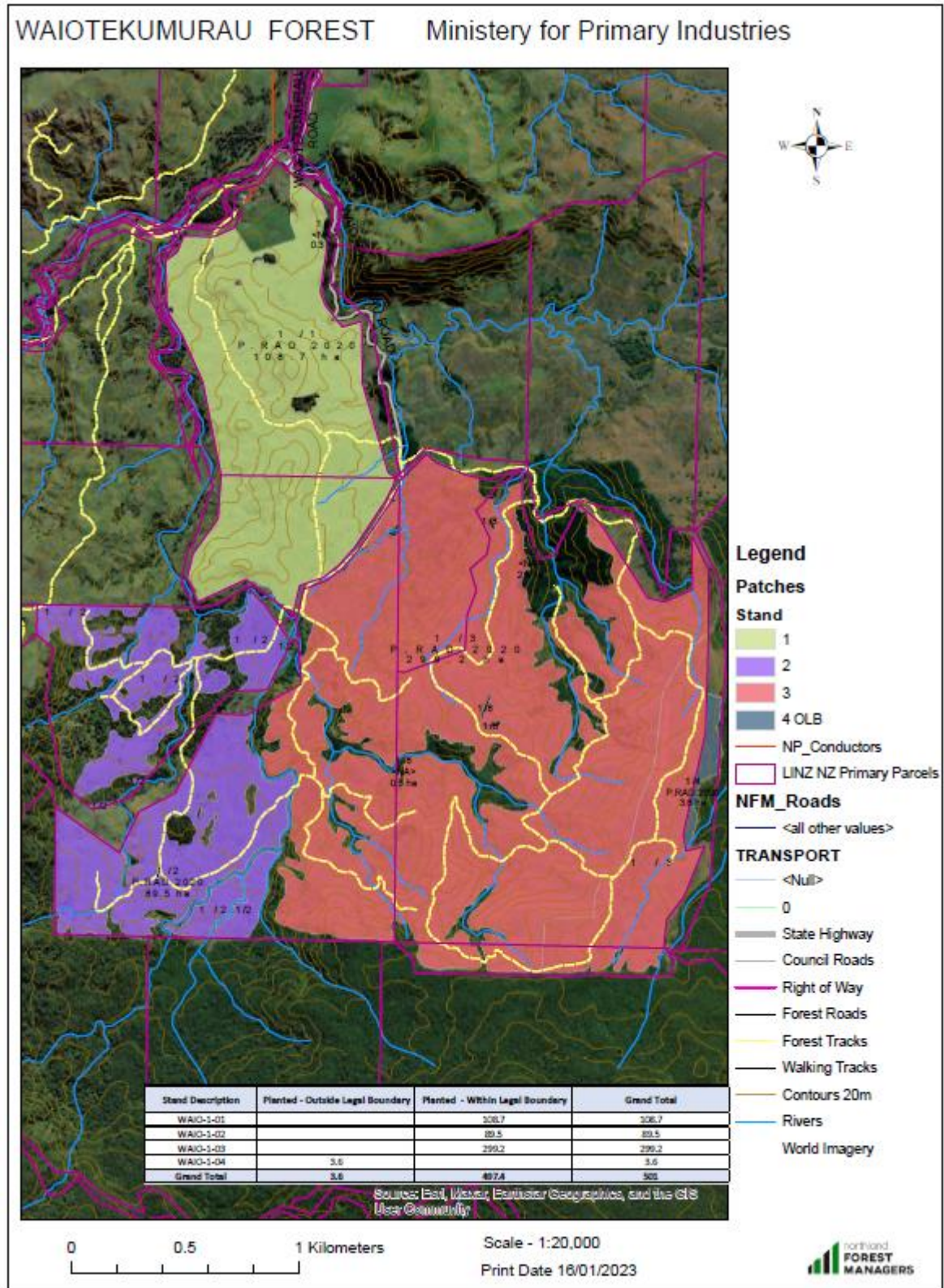
Other relevant legislation in relation to the growing and harvesting of the tree crop are:

- | | |
|---|------|
| • Forests Act | 1949 |
| • Bio-security Act | 1993 |
| • Fencing Act | 1978 |
| • Fire and Emergency New Zealand Act | 2017 |
| • Heritage New Zealand Pouhere Taonga Act | 2014 |
| • Land Drainage Act | 1908 |
| • Accident Compensation Act | 2001 |
| • Civil Aviation Act | 1990 |

Professional management requires all personnel adhere to all relevant legislation. This is important to effectively manage the forest, and to fulfil legal obligations.

Key Work Programme 2024 - 2029	Task	Timing	Key Responsibility
	Review this Plan	Five yearly - next update 2029	MPI
	Prepare Annual Operations report	Annually in February	MPI
	Grantor/Grantee meeting	Annually in February	MPI/Grantor
	Monitor forest health	Annually	MPI
	Dothistroma spray	2024-2026	MPI
	Fertilising	2024-2025	MPI
	Silvicultural inventory	2024-2028	MPI
	Pruning	2024 -2029	MPI
	Waste thinning	2026-2029	MPI
	Road maintenance	Ongoing	MPI
	Forest protection, maintenance work, rates, and levy payments.	Ongoing	MPI

Appendix 1 – Forest Map



Appendix 2 - Forestry Right Map

Annexure Schedule

Insert instrument type

Forestry Right – profit à prendre

Dated

Page

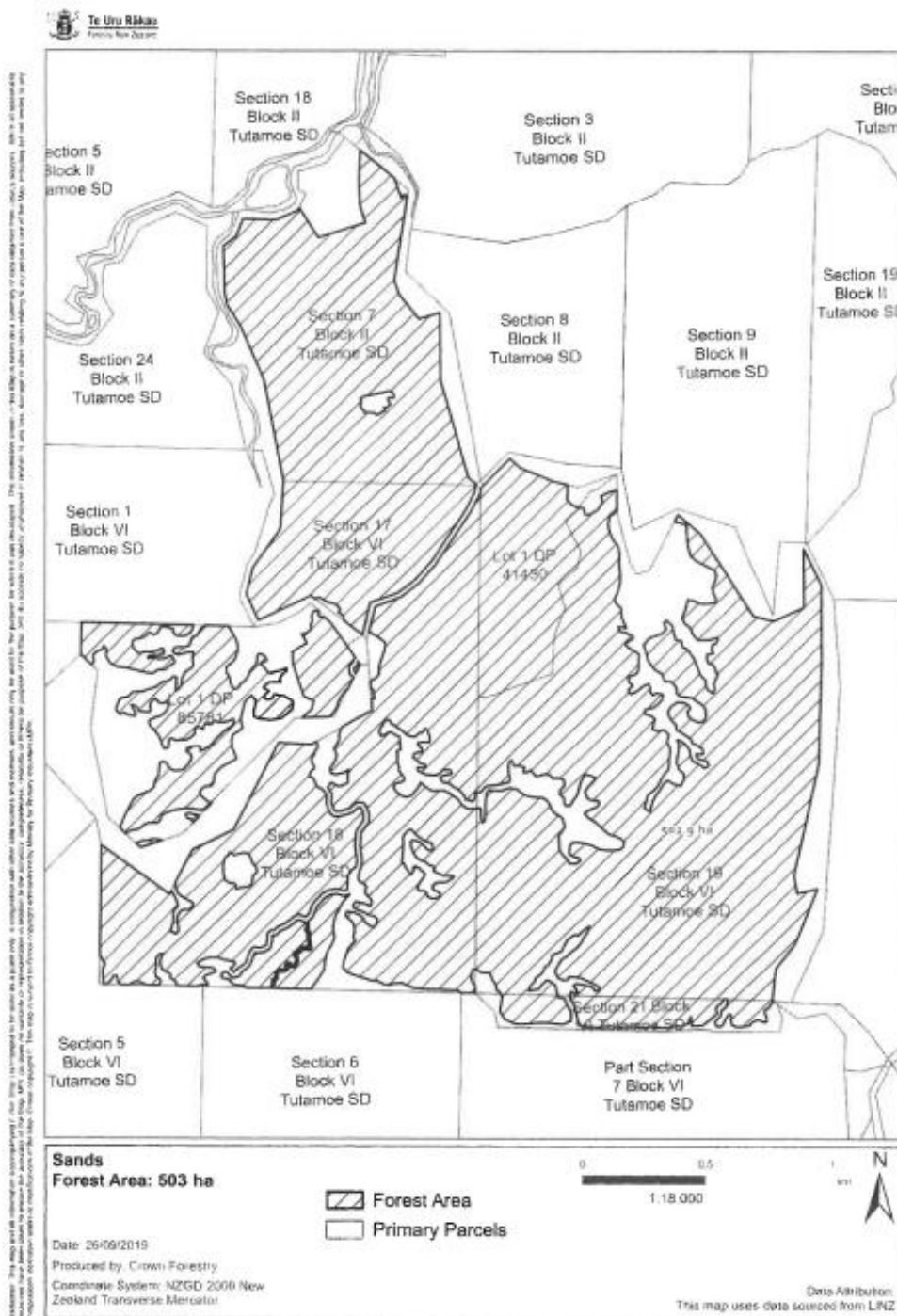
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Pages

Continue in additional Annexure Schedule, if required



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